# Chapter 7. Stress in beam

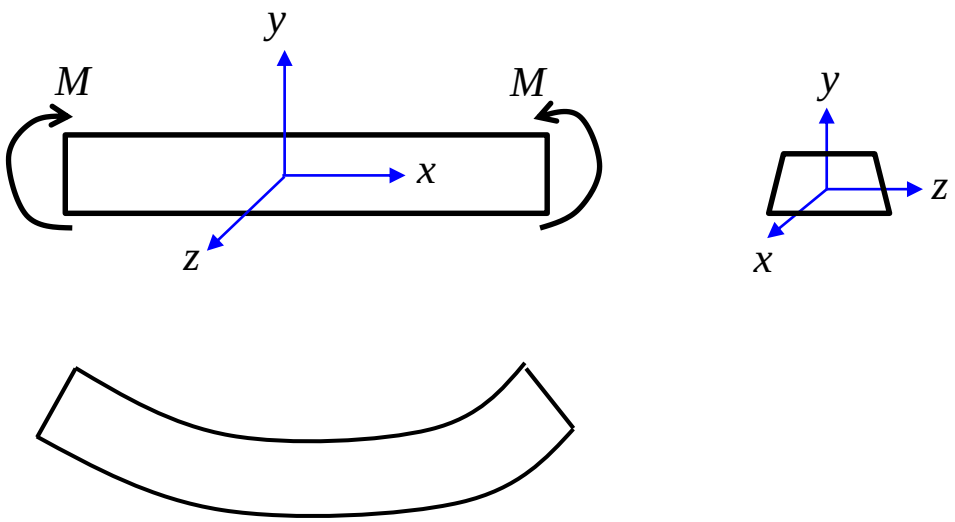
## 7.1. Introduction

Having seen how to determine the internal bending moment in a beam being bent, we now look at how to use this to calculate the strain and stress in the beam.

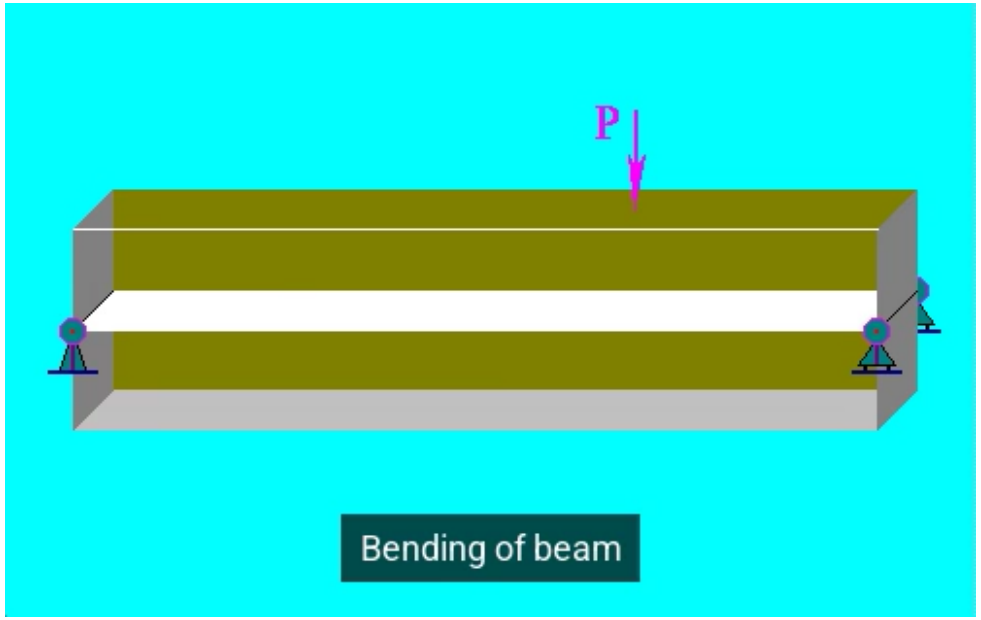
Starting with a beam of length $L$ with a cross-section that is

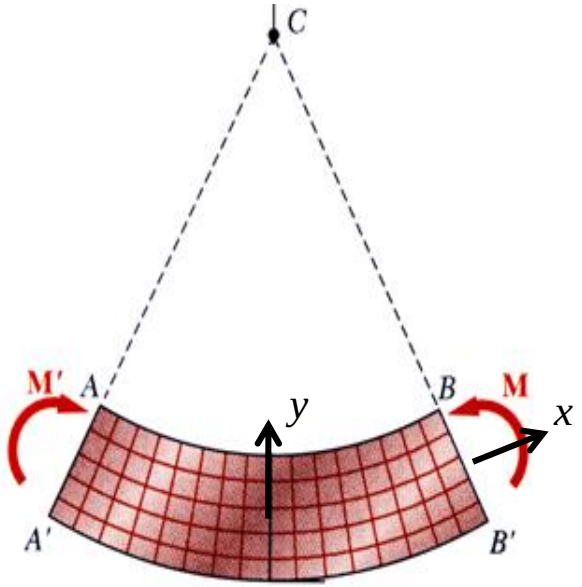
- Constant along the beam (prismatic), and
- Symmetrical about the plane of the page.

Apply a couple $M$ to each end of the beam, causing the beam to bend.

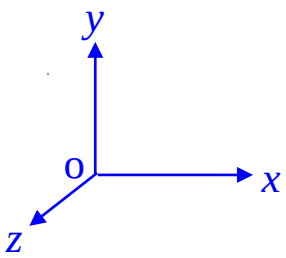


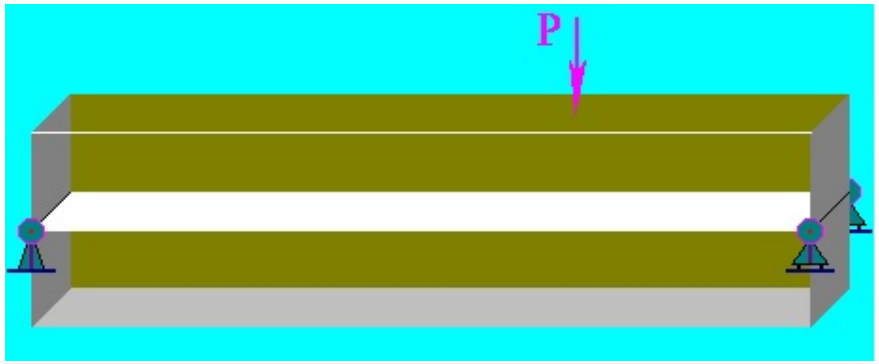
## 7.2. Plane assumption



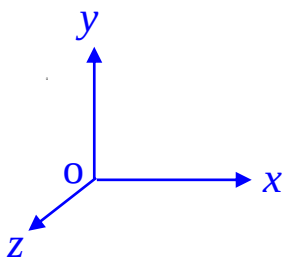



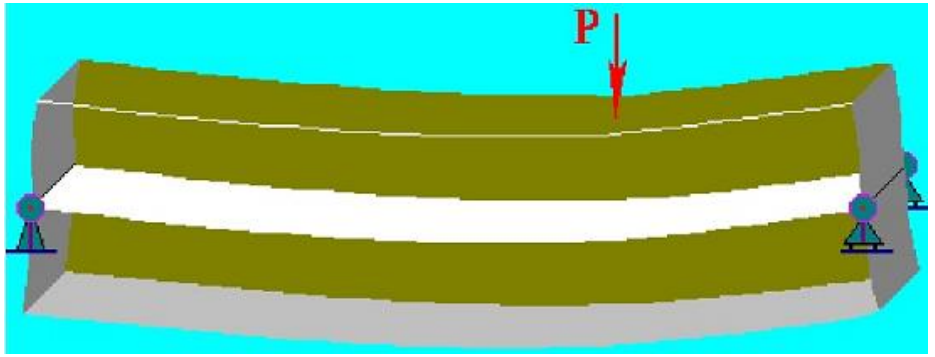
**We can see that:** there is no obvious displacement normal to the plane of the page




Original beam




Deformed beam

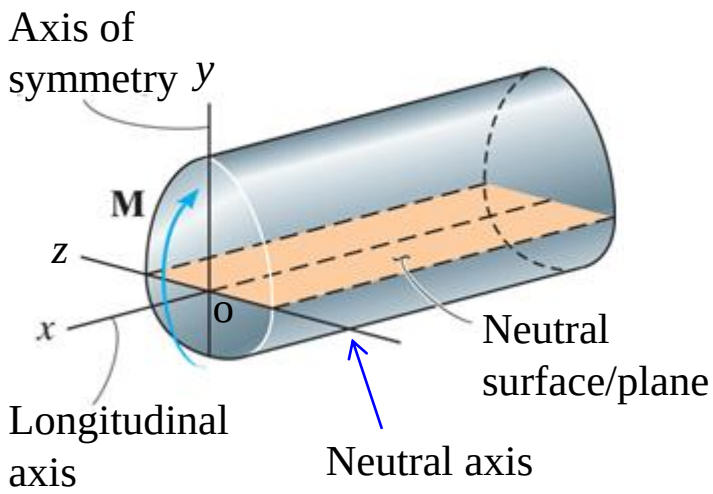
## 7.3. Neutral plane, neutral axis

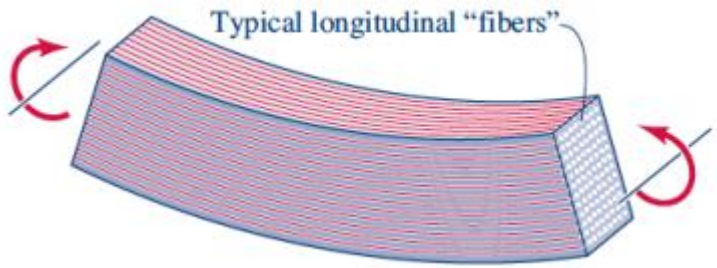
Under positive bending moment $M$ :

➢Top surface under compression

➢Bottom surface under tension

➢A surface in between has *no length change*, and *no stress*, which is called **Neutral plane or Neutral surface.**



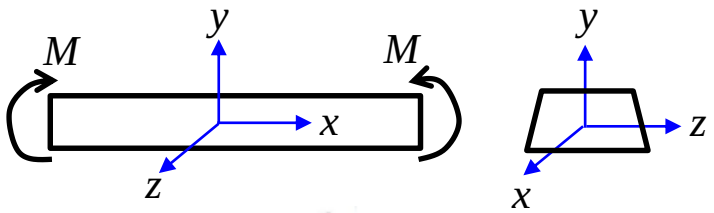

The coordinate axes are arranged such that:

(1) the $x$ axis is along the beam,

(2) the $xoz$ plane is the neutral surface/neutral plane,

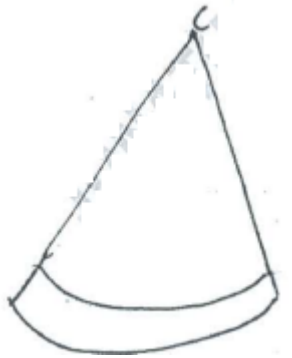
(3) the $z$ axis is the **neutral axis**.

## 7.4. Strain under bending



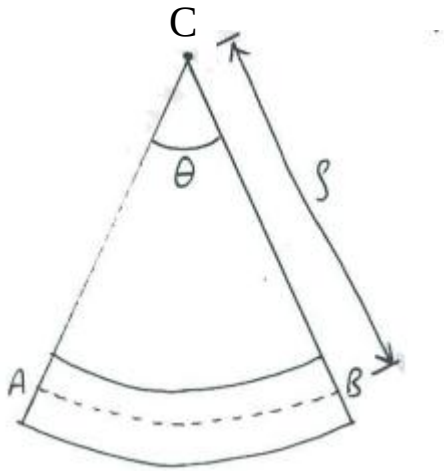

The strain of a beam being bent can be determined from the shape of the arc it bends into.

Provided the cross-section of the beam is symmetrical about the plane of the page and the couples are in the plane of the page, the beam will stay in the plane of the page as it bends.




As *M* is constant along the beam, the beam will bend into an even arc. Call the point at the centre of the arc C.

The bottom face gets longer and the top face gets shorter.



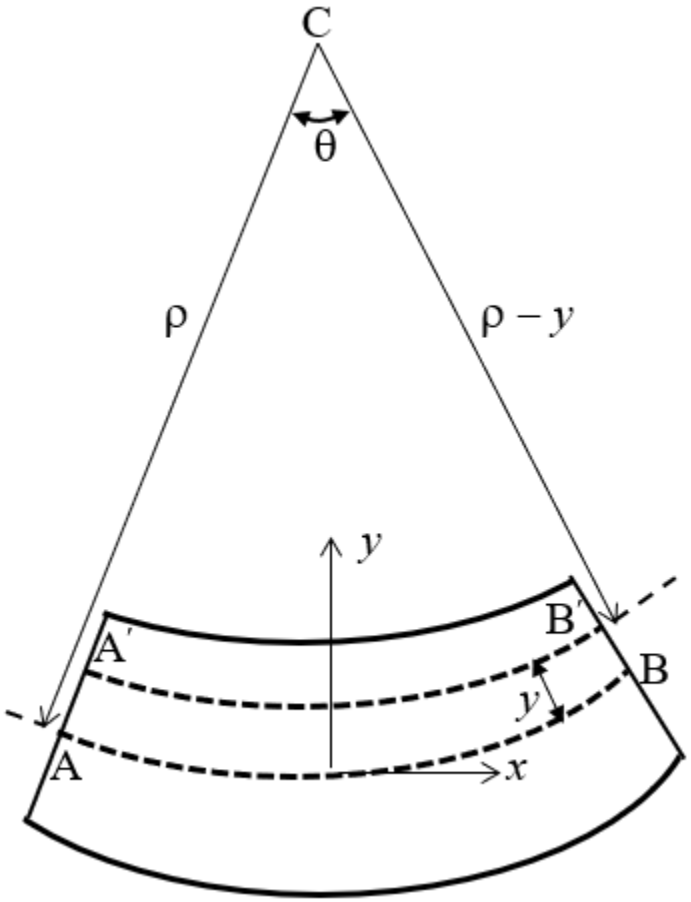

There is a plane somewhere between the top and bottom faces that neither gets longer nor gets shorter. We label this plane AB and call it the Neutral Plane.

The neutral plane's (AB) length remains equal to the original length of the beam (*L*).

$L_{AB} = L$

We define the following dimensions:

- Radius of curvature of the neutral plane, ρ.
- The angle that the beam makes at its centre of curvature, θ




$L_{AB} = L$

$\kappa = 1/\rho$ is the curvature of the **bent** beam.
ρ: radius of curvature. It is the radius of the arc.

Considering the neutral plane (AB) and a second plane (A′B′), a distance $y$ above the neutral plane ($y$ is positive toward the centre of curvature and $y = 0$ is at the neutral plane). Denote the elongation of A′B′ as δ. Note that δ will be negative when $y$ is positive, because A′B′ gets shorter.

We can relate the angle θ to the arcs AB and A′B′.

On AB : $\theta = \dfrac{L}{\rho}$ On A′B′ : $\theta = \dfrac{\overline{A'B'}}{\rho - y} = \dfrac{L + \delta}{\rho - y}$

Equating gives: $\dfrac{L}{\rho} = \dfrac{L + \delta}{\rho - y}$

$$\delta = \frac{-Ly}{\rho} = -L\kappa y$$

This is the elongation of the arc at $y$

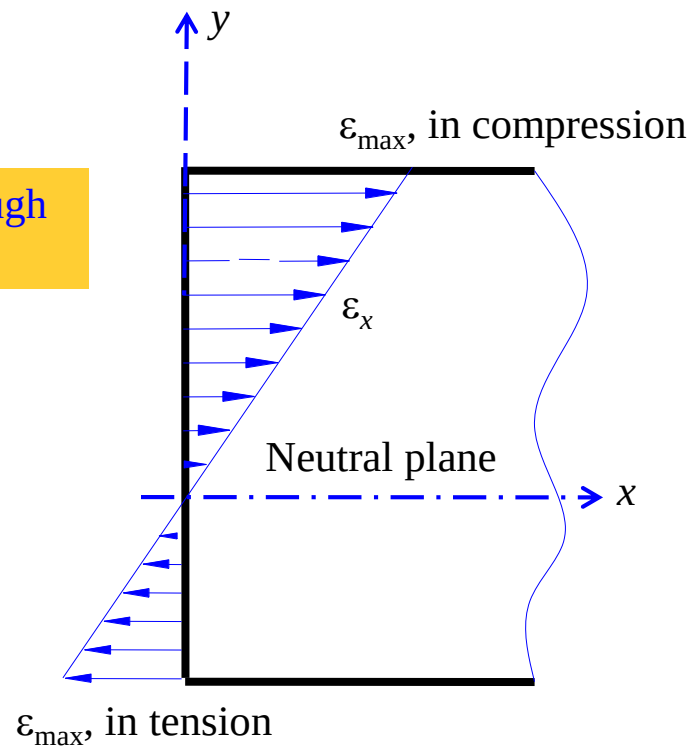
$$\varepsilon_x = \frac{\delta}{L} = -\frac{y}{\rho} = -\kappa y \qquad (7.1)$$

This is the normal stain of the arc at $y$. The strain is in direction $x$.

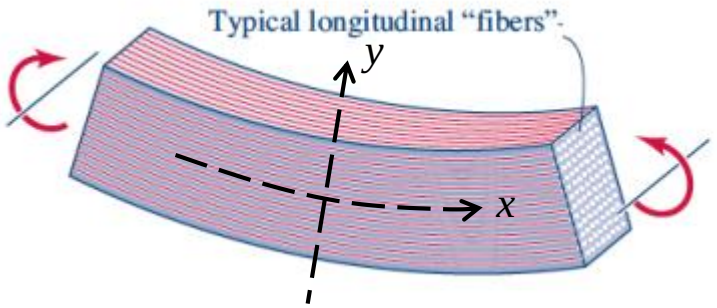
The subscript $x$ is introduced to indicate that the strain is along the axial direction of the beam.

We see that the strain varies linearly with distance from the neutral plane (the $y = 0$ plane).

Linear variation of strain through the thickness of the beam




- Above the neutral plane ($y > 0$), the strain is negative (compressive);
- Below the neutral plane ($y < 0$), the strain is positive (tensile);
- At the neutral plane, $y = 0$, $\varepsilon_x = 0$;
- Maximum strain occurs at the surface (top or bottom) which is farthest away from the neutral plane



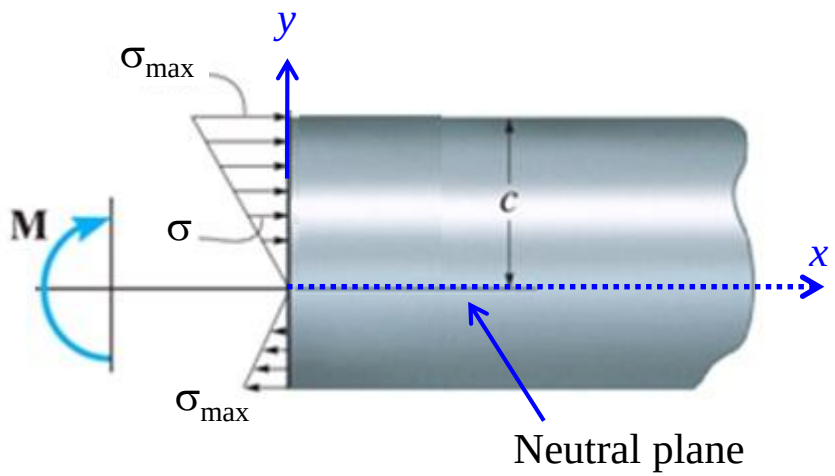
## 7.5. Stresses in a beam under bending

We now use equilibrium to relate the internal stresses to the internal bending moment in the beam.

Assume the material of the beam obeys Hooke's Law. The stress is obtained as

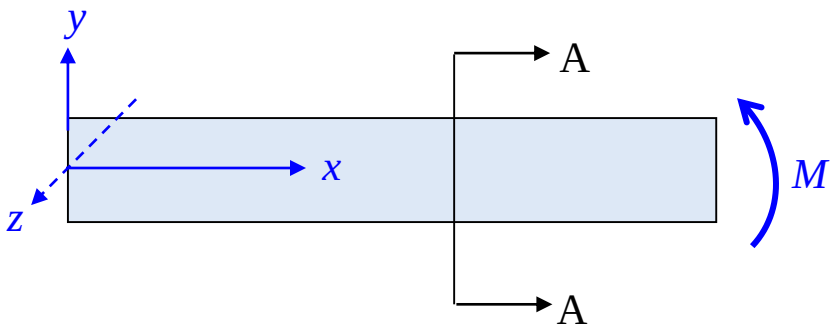
$$\sigma_x = E\varepsilon_x = -\frac{Ey}{\rho} = -E\kappa y \qquad (7.2)$$

- Maximum normal compressive stress occurs at the top surface of beam;
- Maximum normal tensile stress occurs at the bottom surface of beam;
- At the neutral plane, stress is zero.



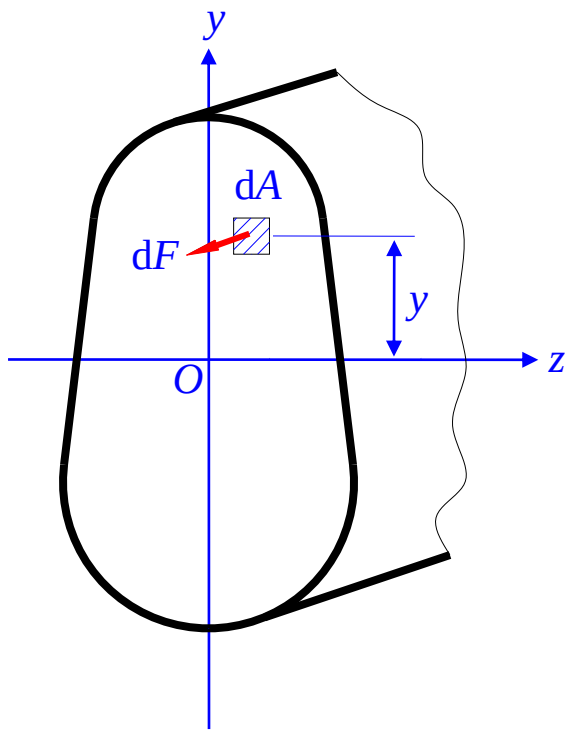

Linear variation of stress through the thickness of the beam

## 7.6. Relating the bending stress to the bending moment in beam




The internal bending moment is defined and given by:

$$M = -\int_A \mathrm{d}F \times y = -\int_A (\sigma_x \mathrm{d}A) y = -\int_A \sigma_x y \mathrm{d}A \qquad (7.3)$$




A-A section (size not in proportion)

By substituting Eq. (7.2) into Eq. (7.3), we get the following moment-curvature relationship *

$$M = \int_A E\kappa y^2 \mathrm{d}A = EI\kappa \qquad (7.4)$$

$$I = \int_A y^2 \, \mathrm{d}A$$

• $I$ : Second moment of area (**moment of inertia**) about the neutral axis. It depends not only on the cross-section area, but also on the shape of the cross-section and on the location of the neutral plane. However, it does not depend on the material property of the cross-section.

• For the expressions of the second moment of inertia for common shapes, you can refer to the textbook. In this unit, we will use the results directly.

From Eq. (7.4), the curvature of the beam can be expressed in terms of the internal moment:

$$\kappa = \frac{1}{\rho} = \frac{M}{EI} \qquad (7.5)$$

Substituting this into Eqs. (7.1) and (7.2) yields the stress and strain in terms of the internal moment:

$$\varepsilon_x = -\frac{My}{EI} \qquad (7.6)$$

$$\sigma_x = -\frac{My}{I} \qquad (7.7)$$

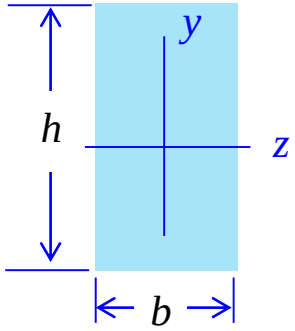
These are known as Engineers' Theory of Bending

The quantity *EI* is called the flexural rigidity of beam.

## 7.7. Moment of inertia

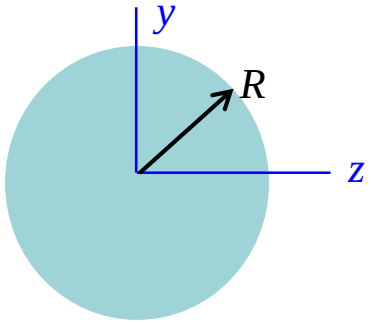
For common shapes, we can look up tables – (e.g., from the textbook)

Rectangular cross-section



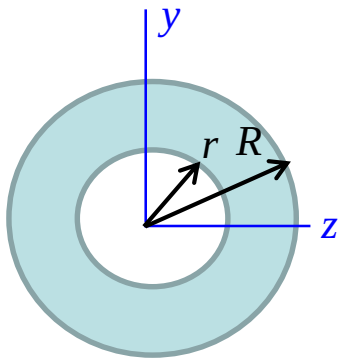

$$I_z = \frac{bh^3}{12} \quad (7.8)$$

Solid circular cross-section



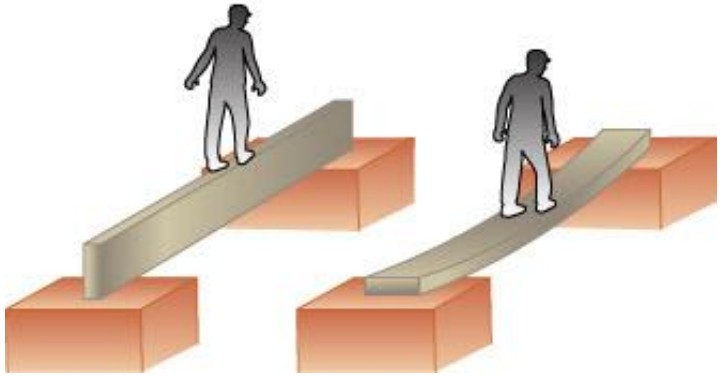

$$I_z = \frac{\pi R^4}{4} \quad (7.9)$$

Circular ring




$$I_z = \frac{\pi R^4}{4} - \frac{\pi r^4}{4} \quad (7.10)$$

Subscript $z$ indicates that the moment of inertia is about the $z$ axis.



(a) (b)

$$I = \frac{bh^3}{12}$$

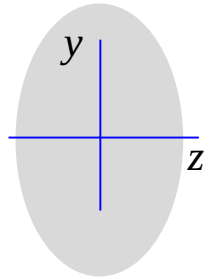
Can you explain the bending behaviour of the same plank when it is orientated differently as shown in the left figure?

Yes, I can hardly bend the beam in (a) and easily bend the beam in (b).

## 7.8. Location of the neutral axis

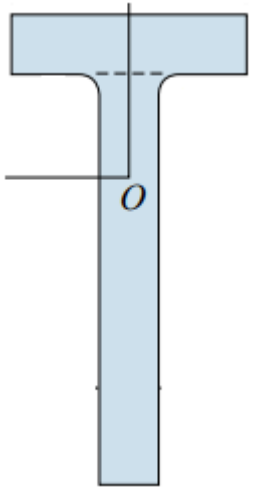
Equations (7.6) and (7.7) express the strain and stress in terms of the moment of inertia of the cross section of the beam about its neutral axis. This is, *y* measures from the neutral axis. Therefore, it is necessary to find the location of the neutral axis.

* For cross-section with regular shape (such as circle or rectangular), its neutral axis passes through the geometry center of the cross-section.

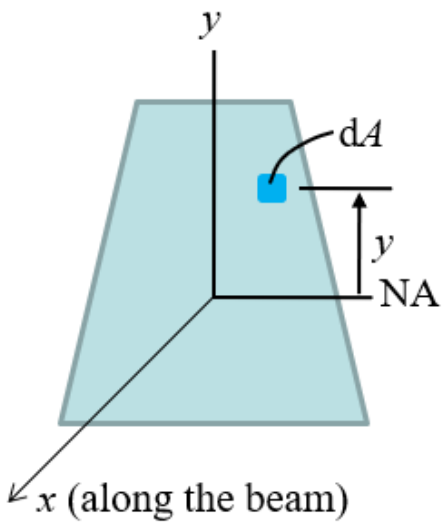



* Symmetric shape, its neutral axis also passes through the center.

Cross-sections with irregular shapes ?



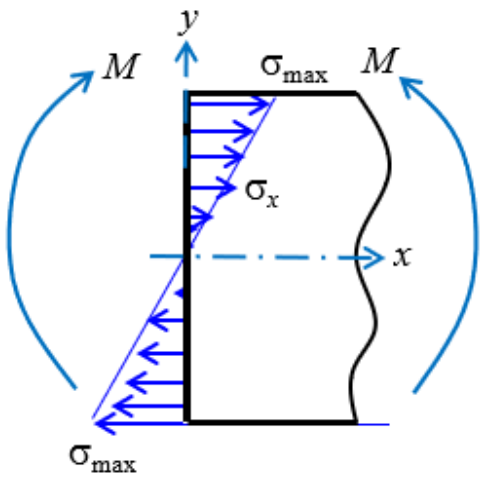
* We consider an element of area d$A$ in the cross-section of the beam. The element is located at distance $y$ from the neutral axis, and therefore the stress $\sigma_x$ acting on the element is given by $\sigma_x = -My/I$. The *force* acting on the element is equal to $\sigma_x\,dA$ and is compressive when $y$ is positive. Because there is no resultant force acting on the entire cross section, the integral of $\sigma_x dA$ over the area of the entire cross section must vanish:




$$\int_A \sigma_x dA = -\int_A \frac{My}{I} dA = 0$$

* Because $M$, and $I$ are nonzero constants at any given cross section of a bent beam, they are not involved in the integration over the cross-sectional area. Therefore, we can drop them from the equation and obtain

$$\int_A y dA = 0$$

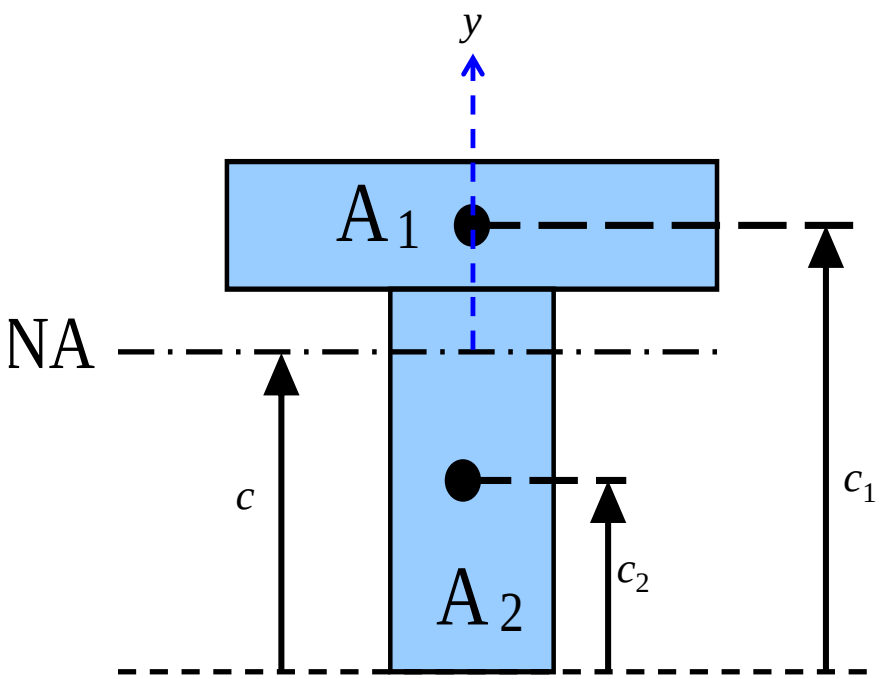



* This equation states that the neutral axis axis must pass through the **centroid of the cross section**. Therefore, we obtain the following important conclusion:

*The neutral axis passes through the centroid of the cross-sectional area* when the material follows Hooke's law.

This observation makes it relatively simple to determine the position of the neutral axis.

## Centroid of composite shapes




$y_1 = c_1 - c$

$y_2 = c_2 - c$

The definition of the centroid of a composite shape is

$$\int_A y\mathrm{d}A = y_1A_1 + y_2A_2 = 0$$

$$\Rightarrow \quad (c_1 - c)A_1 + (c_1 - c)A_1 = 0$$

$$\Rightarrow \quad (A_1 + A_2)c = c_1A_1 + c_2A_2$$

$$\Rightarrow \quad c = \frac{c_1A_1 + c_2A_2}{A_1 + A_2} \qquad (7.11)$$

Where

- $c_1$, $c_2$ etc are the distance to the centre of each sub-shape.
- $c$ is the distances up to the neutral plane of the combined shape.
- $A_1$, $A_2$ etc are the areas of each sub-shape.

For centroids of common shapes, we can look up tables in the textbook(s).

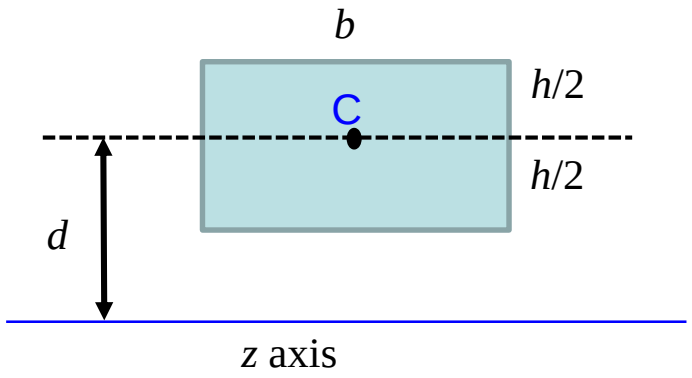
## 7.9. Parallel - axis theorem

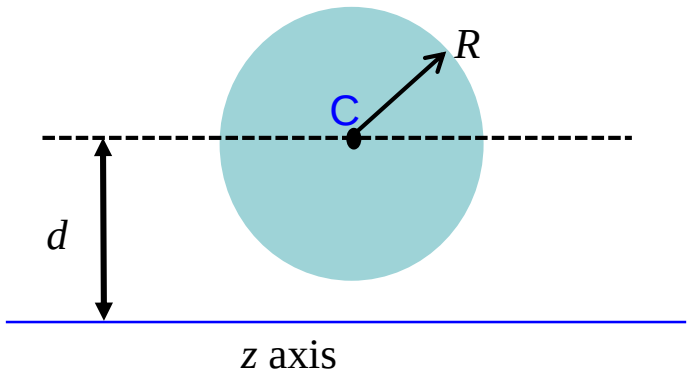
**Parallel-Axis Theorem :** The second moment of an area about an axis (e.g., the reference line of below figure) is equal to the area's second moment about a parallel axis passing through the "centroid" plus the product of the area and the square of the perpendicular distance between the axes:

$$I_z = I_c + Ad^2 \qquad (7.12)$$

You have seen this equation in the course *Fundamentals of Mechanics*.

where $A$ is the area of the shape, $d$ is the distance from the centroidal axis of the shape to the desired parallel axis (e.g, the $z$ axis), $I_c$ is the second moment of the area about its own centroid, which has a distance $d$ from the $z$ axis.



## 7.10. Composite shapes (same materials but different shapes)

**(1)** Consider the shape as the combination of some simple/regular elements: $A_1$, $A_2$,….

**(2)** Determine the location of neutral axis (based on Eq. 7.11):

$$c = \frac{\sum_{i=1}^{N} c_i A_i}{\sum_{i=1}^{N} A_i} \qquad (7.11)$$

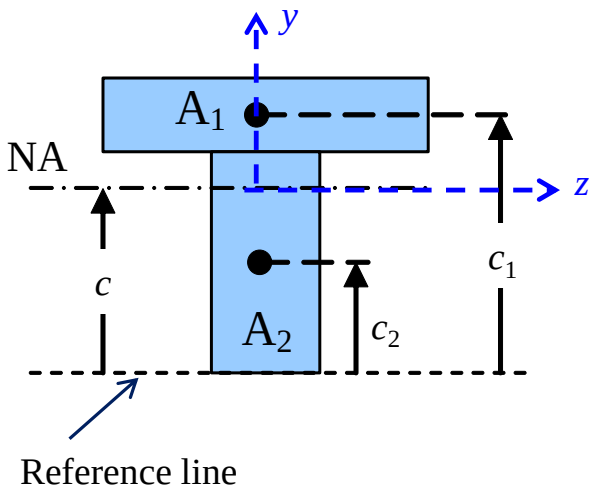
**(3)** Calculate $I_i$: the second moment of each simple element about the $z$ axis (the neutral axis, NA) by parallel axis theorem:

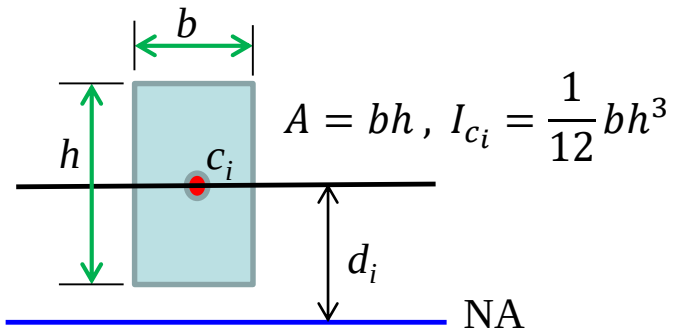
$$I_{zi} = I_{c_i} + A_i d_i^2 = I_{c_i} + A_i (c_i - c)^2 \qquad (7.12)$$

This is the moment of the *i*-th simple element about its own neutral axis (e.g, the central point $c_i$ of the below rectangular element).

**(4)** Second moment of the shape is the sum of all simple/regular elements:

$$I_{\text{total about NA}} = I_1 + I_2 + \cdots$$



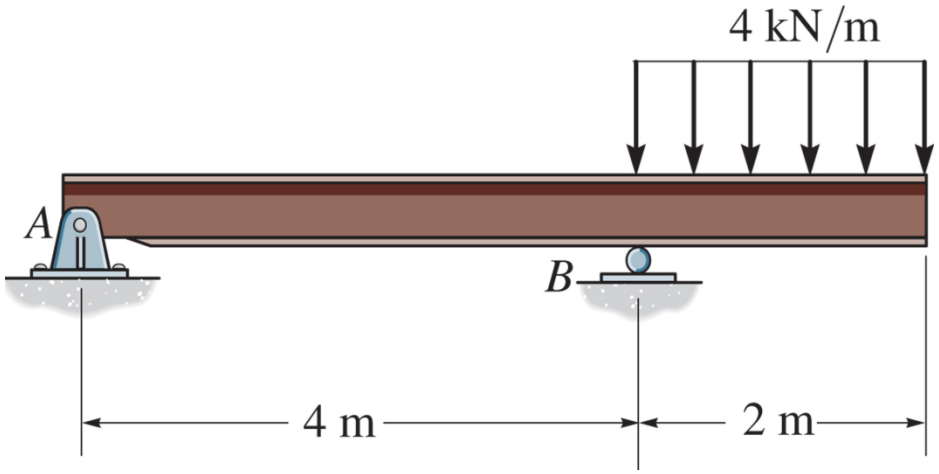

**(5)** Calculate the bending stress from:

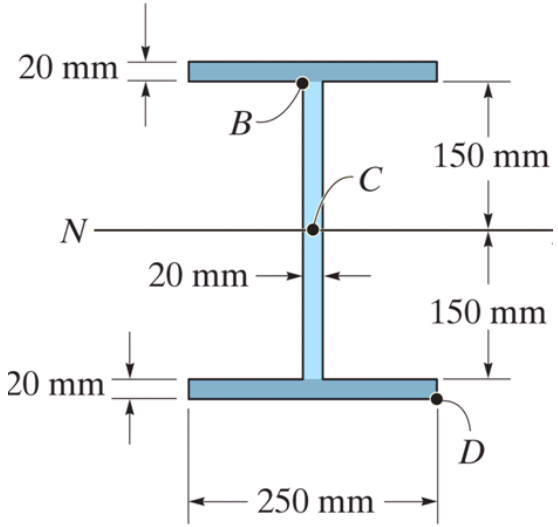
$$\sigma = -\frac{My}{I}$$

In which $y$ is where you want to find the stress. It is measured from the $z$ axis (the NA of the cross-section).

**Example 1.** The simply supported beam has the *I*-shape cross-sectional area as shown.

Determine the maximum bending stress in the beam



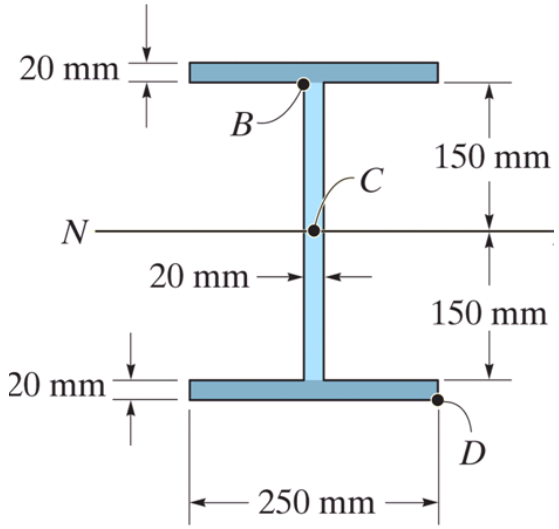

**Step 1:** Find the **maximum** bending moment in the beam (for this problem, obviously, its at point B):

$$M_{\max} = -(4{,}000 \times 2) \times \frac{2}{2} = -8{,}000 \text{ N.m}$$

**Step 2**: Section property: second moment of area

By symmetry, the neutral axis passes through the centroid $C$ that is at the mid-height of the beam. The second moment of area is

$$I = \sum\left(\bar{I} + Ad^2\right)$$
$$= 2\left[\frac{1}{12}(0.25)(0.02)^3 + (0.25)(0.02)(0.16)^2\right] + \left[\frac{1}{12}(0.02)(0.3)^3\right]$$
$$= 301.3\left(10^{-6}\right)\ \text{m}^4$$

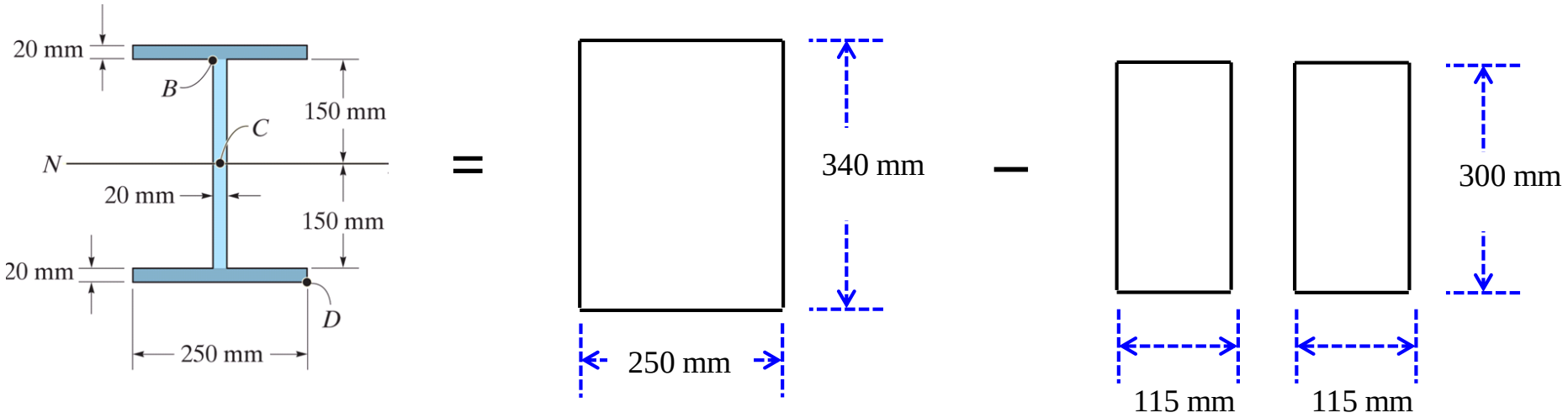



**Step 3:** Applying the flexure formula where $y_{\max} = \pm 170$ mm,

$$\sigma_{\max} = -\frac{My_{\max}}{I} = -\frac{-8\times10^3(\pm0.17)}{301.3\times10^{-6}} = \pm4.51\times10^6\ \text{Pa} = \pm4.51\ \text{MPa}$$

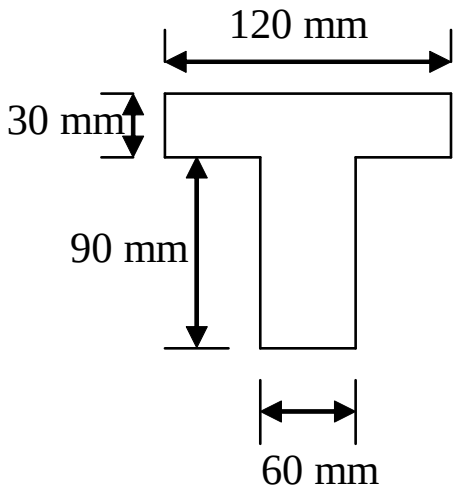
**Note:** **Another way to find *I***

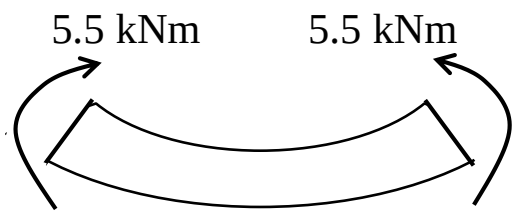
Image the cross section is formed by subtracting a small cross-section (230 mm × 300 mm) from a big cross-section (250 mm × 340 mm)




$$I = \frac{250 \times (300 + 40)^3}{12} - \frac{(250 - 20) \times 300^3}{12} = 301.3 \times 10^6 \text{ mm}^4 = 301.3 \times 10^{-6} \text{ m}^4$$

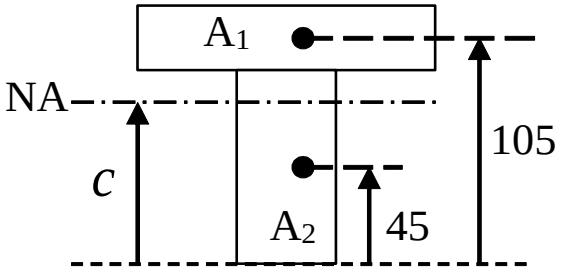
**Example 2.** Find the maximum bending stress




$M$ = 5,500 N•m

= 5,500×1,000 N•mm




**Solution:**

**Step 1:** Split the cross-section's shape into two simple elements, A1 and A2 as shown

$$A_1 = 120 \times 30 = 3600 \text{ mm}^2$$

$$A_2 = 90 \times 60 = 5400 \text{ mm}^2$$

$$c_1 = (30/2 + 90) = 105 \text{ mm}$$

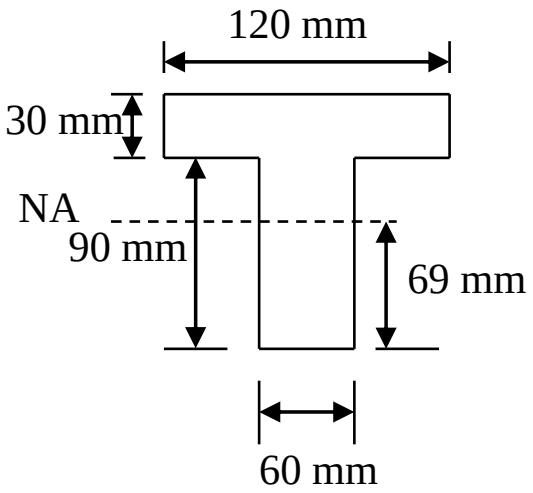
$$c_2 = 90/2 = 45 \text{ mm}$$

**Step 2:** Locate the neutral Axis (NA)

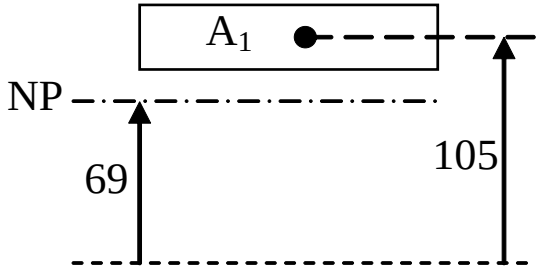
$$c = \frac{c_1 A_1 + c_2 A_2}{A_1 + A_2} = \frac{105 \times 3600 + 45 \times 5400}{3600 + 5400} = 69 \text{ mm}$$

**Step 3:** Calculate the *I* values of each element about NA:

$$I_i = \frac{1}{12} b_i h_i^3 + A_i d_i^2$$

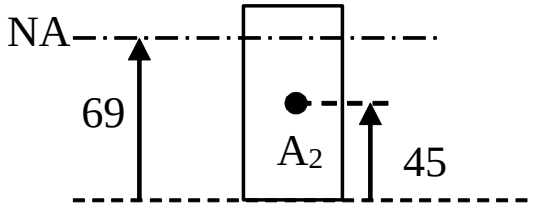



=




$$I_2 = \frac{1}{12} \times 60 \times 90^3 + 5400(69-45)^2$$
$$= 6755400\,\text{mm}^4$$

+




$$I_2 = \frac{1}{12} \times 60 \times 90^3 + 5400(69-45)^2$$
$$= 6755400\,\text{mm}^4$$

**Step 4:** Calculate the total *I* of the composite areas about NA: $I = I_1 + I_2 = 11.691 \times 10^6 \text{mm}^4$.

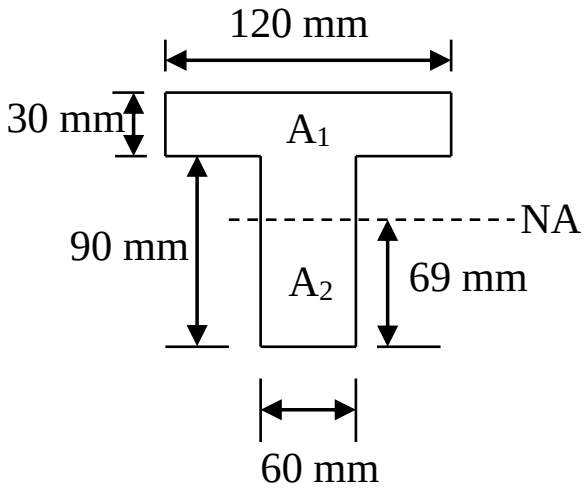
**Step 5:** Calculate the corresponding maximum value of bending stress

Maximum stress in A1 is at the top of the beam:

$$\sigma_{\text{top}} = -\frac{My_{\text{top}}}{I} = -\frac{(5{,}500 \times 1{,}000) \times (90 + 30 - 69)}{11.691 \times 10^6}$$

$$= -23.993 \text{ MPa}$$

compressive




Maximum stress in A2 is at the bottom of the beam:

$$\sigma_{\text{bottom}} = -\frac{My_{\text{bottom}}}{I} = -\frac{(5{,}500 \times 1{,}000) \times (-69)}{11.691 \times 10^6} = 32.461 \text{ MPa}$$

tensile

**Summary:** **Determine the bending stress in the beam**

**Solution Procedure:**

1) Find the support reaction - Chapter 8

2) Find the bending moment at the point where the bending stress is calculated ($M_{max}$ to calculate maximum bending stress) - Chapter 8

3) Determine the location of the neutral axis (centroid of the cross-section of the beam)

4) Calculate the Moment of Inertia of the cross-section ($I$)

5) Calculate the bending stress using the flexure formula : $$\sigma_x = -\frac{My}{I}$$

**Note:** a) if the moment is given, neglect Steps 1 and 2.

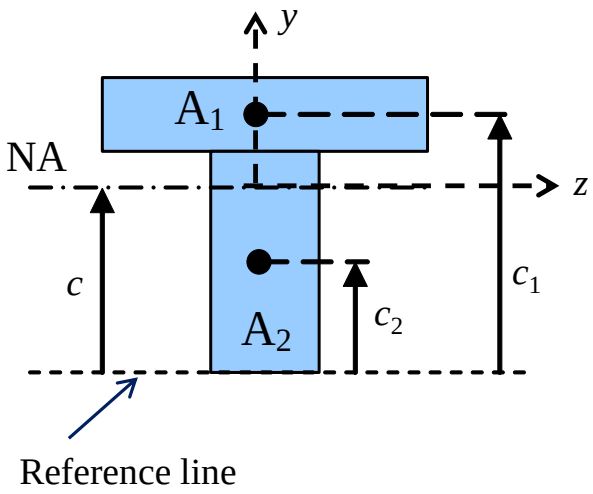
b) If the position of neutral axis is given or known (e.g., for a symmetric cross section), neglect Step 3.

# Quick review: Bending strain and stress

**Normal stress**

$$\sigma_x = -\frac{My}{I}$$

- $E$ is the Young's modulus of the material of the beam.
- $I$ is the moment of inertia of the cross-section of the beam about NA (the $z$ axis)..
- $y$ is the distance at which we want to calculate the stress. $y$ starts from the NA (the $z$ axis).




Curvature (κ) and radius of curvature (ρ) of bent beam:

$$\kappa = \frac{1}{\rho} = \frac{M}{EI}$$

**Location of neutral axis** $c = \dfrac{\sum_{i=1}^{N} c_i A_i}{\sum_{i=1}^{N} A_i}$

**Moment of inertia**

$$I_Z = I_{z1} + I_{z2} + \cdots$$

where ( $i = 1, 2, \ldots$)

$$I_{zi} = I_{c_i} + A_i d_i^2 = I_{c_i} + A_i (c_i - c)^2$$

